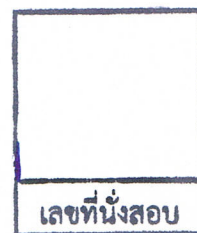


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 2

Subject: 392152 Chemistry 2

Date: 26 December 2013

Year: 2013

Section: 15 -16

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Calculators are allowed in the examination.
4. This exam is a closed book examination.
5. Electronic communication devices are not allowed in the examination room.
6. The examination has 6 pages (including this page), 2 sections and a total score of 50 points.
7. Write all your solutions and answers on this examination sheet.

Relative atomic mass of

H = 1 C = 12 N = 14 O = 16 Na = 23 S = 32 Cl = 35.5

Gas constant; $R = 0.082 \frac{\text{atm.L}}{\text{mol.K}}$

Good luck!!

10. Tyndall effect occurs only in _____.
 a. solution b. suspension c. colloid d. compound
11. The rate of evaporation decrease as _____ increase.
 a. pressure b. temperature c. surface area d. intermolecular force
12. Above sea level, water boils at a _____ temperature due to the _____ atmospheric pressure.
 a. higher, increase b. lower, increase
 c. higher, decrease d. lower, decrease
13. The pressure of a gas in a container is 152 mm Hg. This is equivalent to _____.
 a. 0.2 atm b. 0.3 atm c. 2 atm d. 0.4 atm
14. The normal boiling point of chloroform, which has a higher vapor pressure than water at 100°C, is _____.
 a. higher than the normal boiling point of water b. lower than the normal boiling point of water
 c. the same as the normal boiling point of water d. unable to be measured
15. In which condition that the behavior of a real gas is similar to the behavior of an ideal gas?
 a. high temperature and high pressure b. high temperature and low pressure
 c. low temperature and high pressure d. low temperature and low pressure

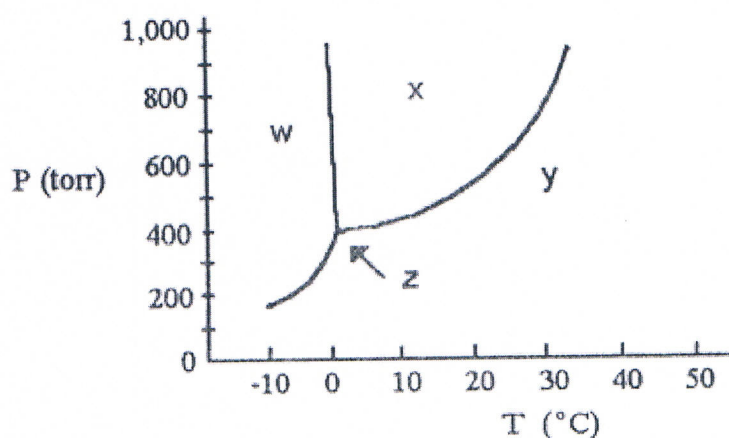
Part 2 Answer (35 points)

1. Fill out the blanks (3 points)

Changing	Changing of phase	Process	Endo/Exothermic
ice→water	solid→liquid	melting	endothermic
water→ice	_____	_____	_____
water→vapour	_____	_____	_____

2. A 2.00-L sample of gas at 675 torr pressure is changed at constant temperature until its pressure is 925 torr. What is its new volume? (2 points)

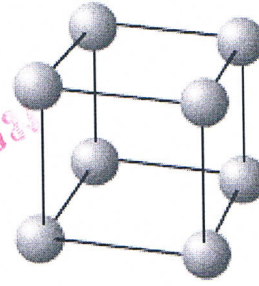
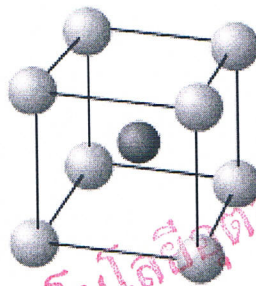
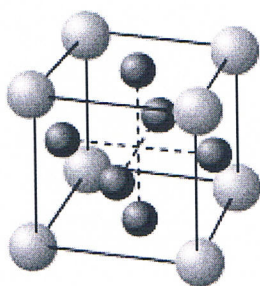
3. A fixed volume container containing hydrogen gas has an initial pressure of 98 kPa at 10 °C. What is the pressure at 25 °C? (2 points)
4. A sample of gas occupies 12.0 liters at 240 °C under the pressure of 80.0 kPa. At what temperature would the gas occupy 15.0 liters if the pressure increased to 107 kPa? (3 points)
5. Find the density (in g/L) of CO₂ at room conditions (20. °C and 1 .00 atm). (3 points)
6. Oxygen gas and sulfur dioxide (SO₂) gas are at the same temperature. What is the ratio of the effusion rates of their molecules? (3 points)



7. According to the phase diagram on the left:
(2 points)

- The normal boiling point is _____.
- The critical point is _____.
- The tripple point is _____.
- The normal phase at 1 atm is _____.

8. Identify the unit cell in each figure below as simple cubic, body-centered cubic, or face-centered cubic (3 points)



8.1 _____

8.2 _____

8.3 _____

9. Complete the following table. (6 points)

	Concentration of a solution	Amount of the solute	Amount of the solvent
9.1	40 %w/w	10 g.	
9.2		2 mol	500 mL
9.3		0.02 g	10 mL
9.4	0.5 M.		5 L.

10. How many grams of H_2O must be used to dissolve 50.0 grams of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) to prepare a 1.25 *m* solution of sucrose? (4 points)

11. A solution is prepared by dissolving 1.8 g of $\text{C}_2\text{H}_{12}\text{O}_6$ in 198 g of water to make 200 mL of solution. Calculate (4 points)

- a) the percentage concentration (%w/w)
- b) the mole fraction of $\text{C}_2\text{H}_{12}\text{O}_6$

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